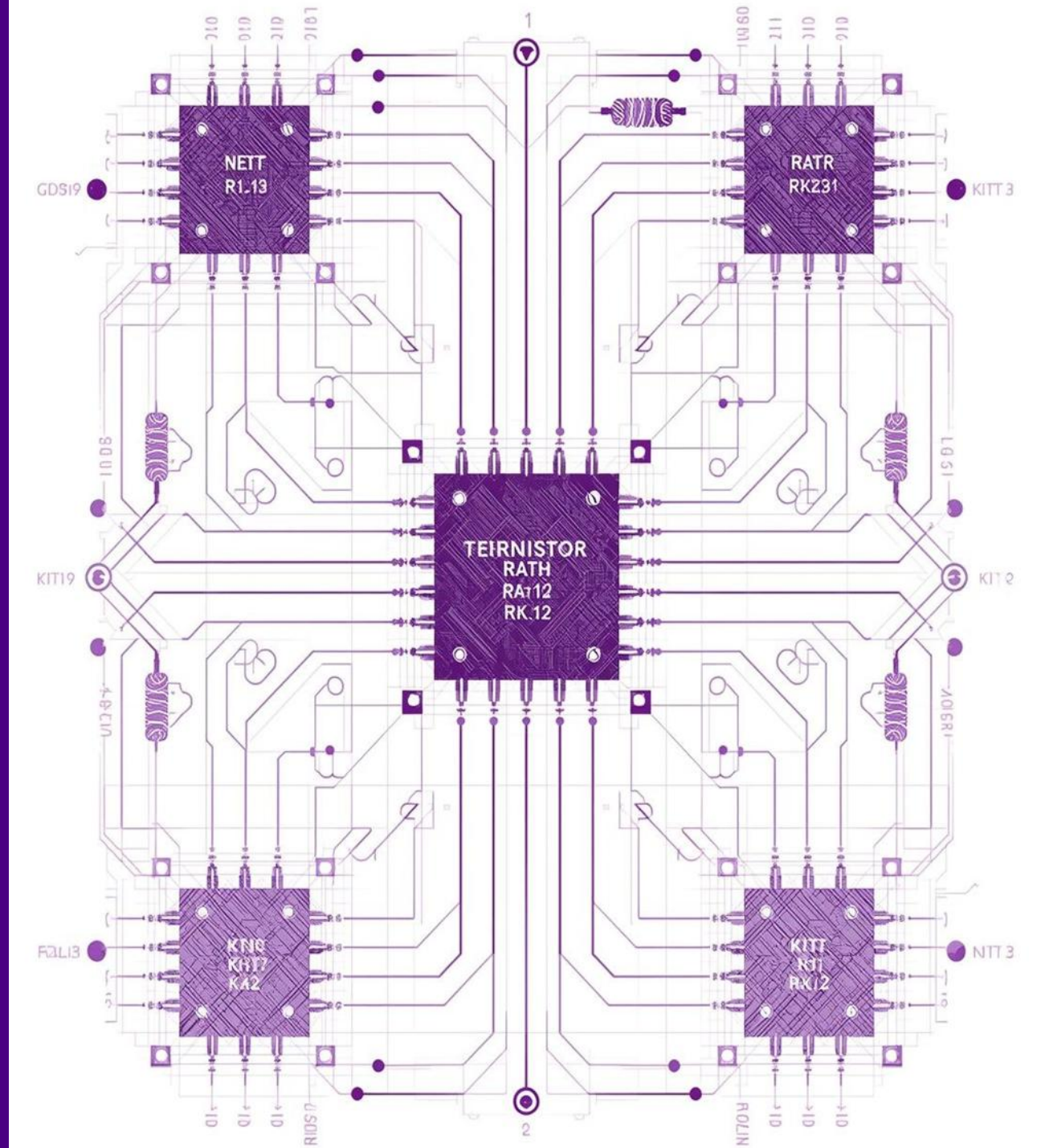


Design and Analysis of a Low Power Single Ended CCFF in 22nm CMOS

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Motivation & Problem Statement

The Power Challenge

- ❑ Clock networks consume up to 45% of total chip power.
- ❑ Goal: Target 20-30% power reduction using a 22nm process.

Motivation & Problem Statement

The Problem: Redundant Switching

- ❑ Conventional flip-flops discharge internal nodes every clock cycle, even if data (D) hasn't changed.
- ❑ Wasted energy on "useless" transitions when $D = Q$

Motivation & Problem Statement

The Solution: Conditional Capture

- ❑ Implement logic to "check" the output before switching.
- ❑ Eliminate internal activity when input matches current state.

Single-Ended CCFF Structure

Operation Principle

CLK = 0: Precharge X_bar to VDD

CLK = 1:

If $D \neq Q \rightarrow$ Discharge X_bar $\rightarrow$ Update output

If $D = Q \rightarrow$ Hold X_bar high $\rightarrow$ **No transition**

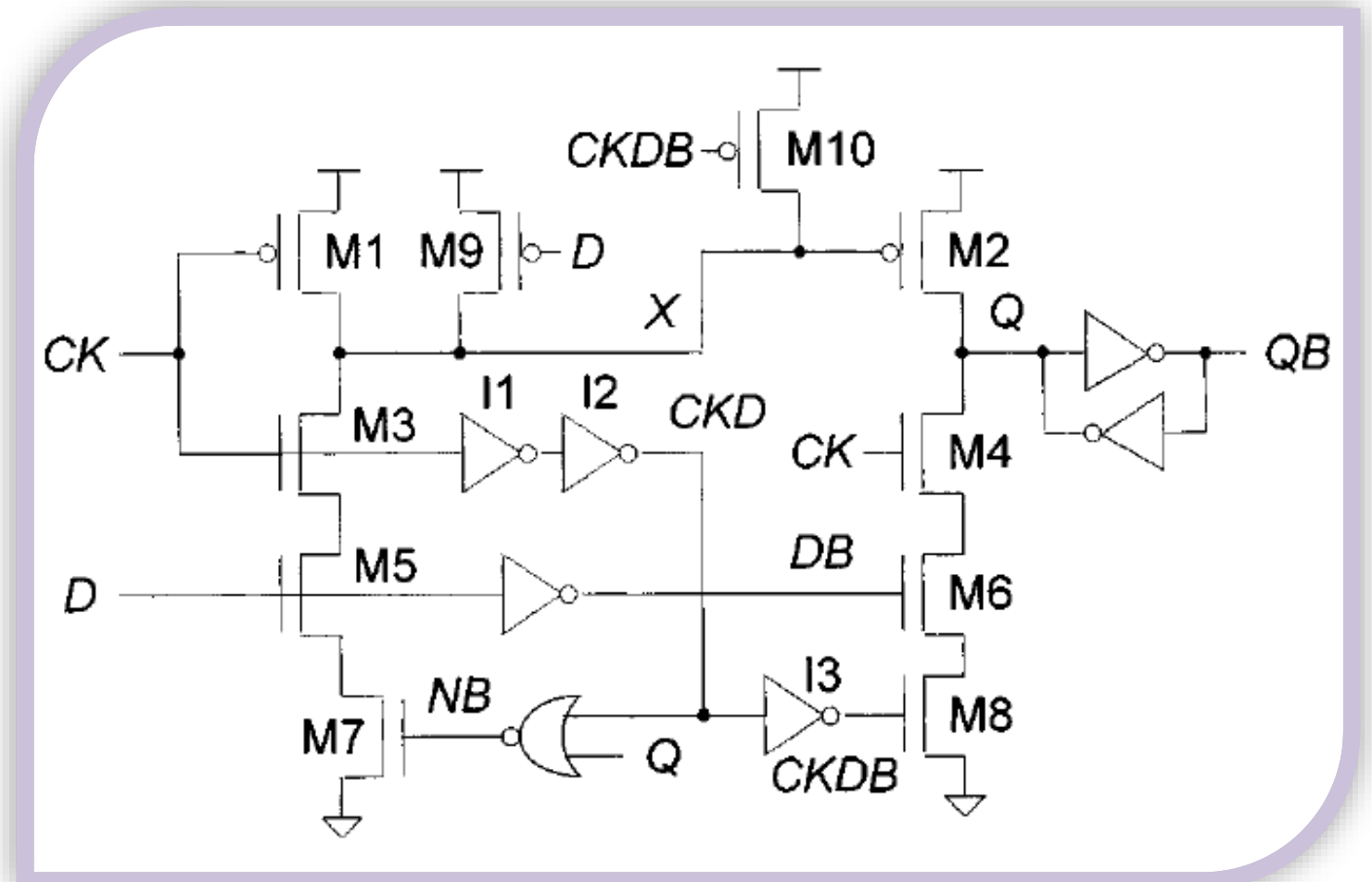
Key Components:

NOR gate: Compares input (D) with output (Q)

Clock inverters: Generate delayed clock signal (CLK_d)

Precharge transistor (M1): Sets X_bar high when CLK = 0

Conditional discharge path: Only activates when $D \neq Q$



Implementation in 22nm CMOS

Technology Specifications

Process: 22nm CMOS technology

Supply Voltage: 0.95V

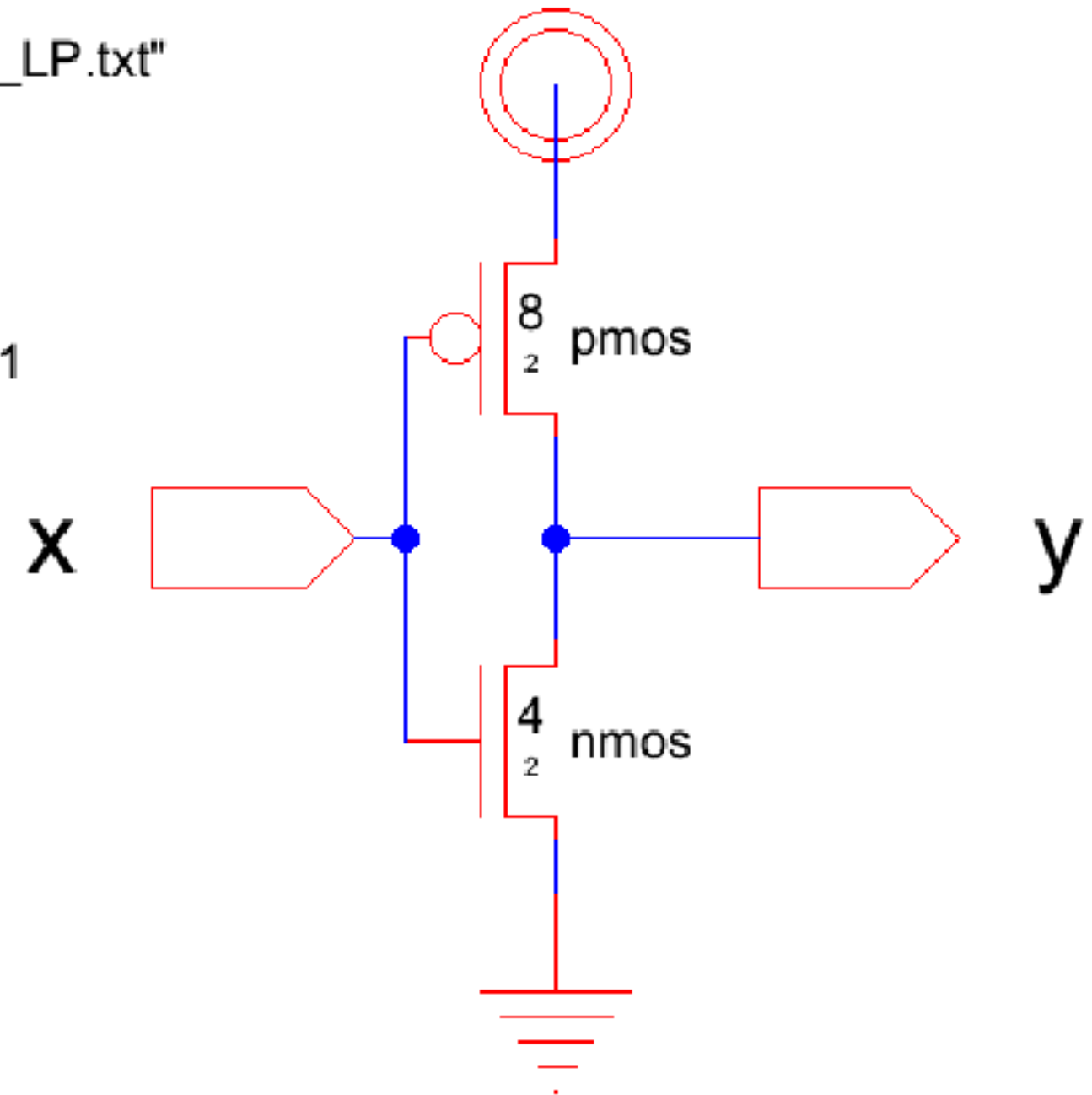
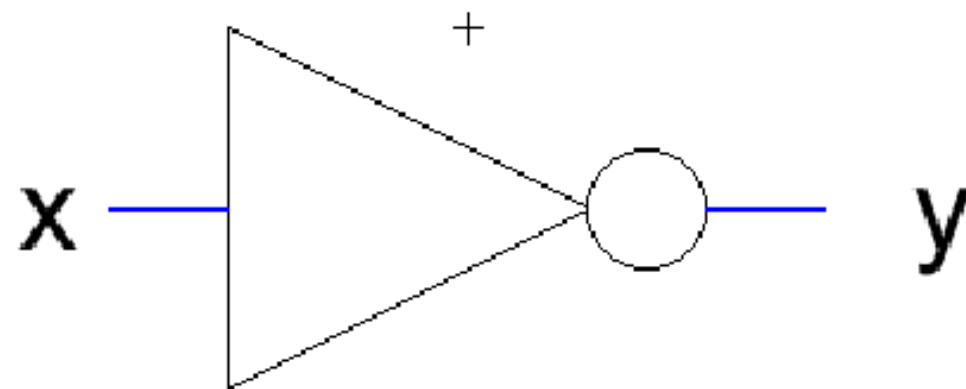
Load Capacitance: 5fF

Transistor Sizing Strategy

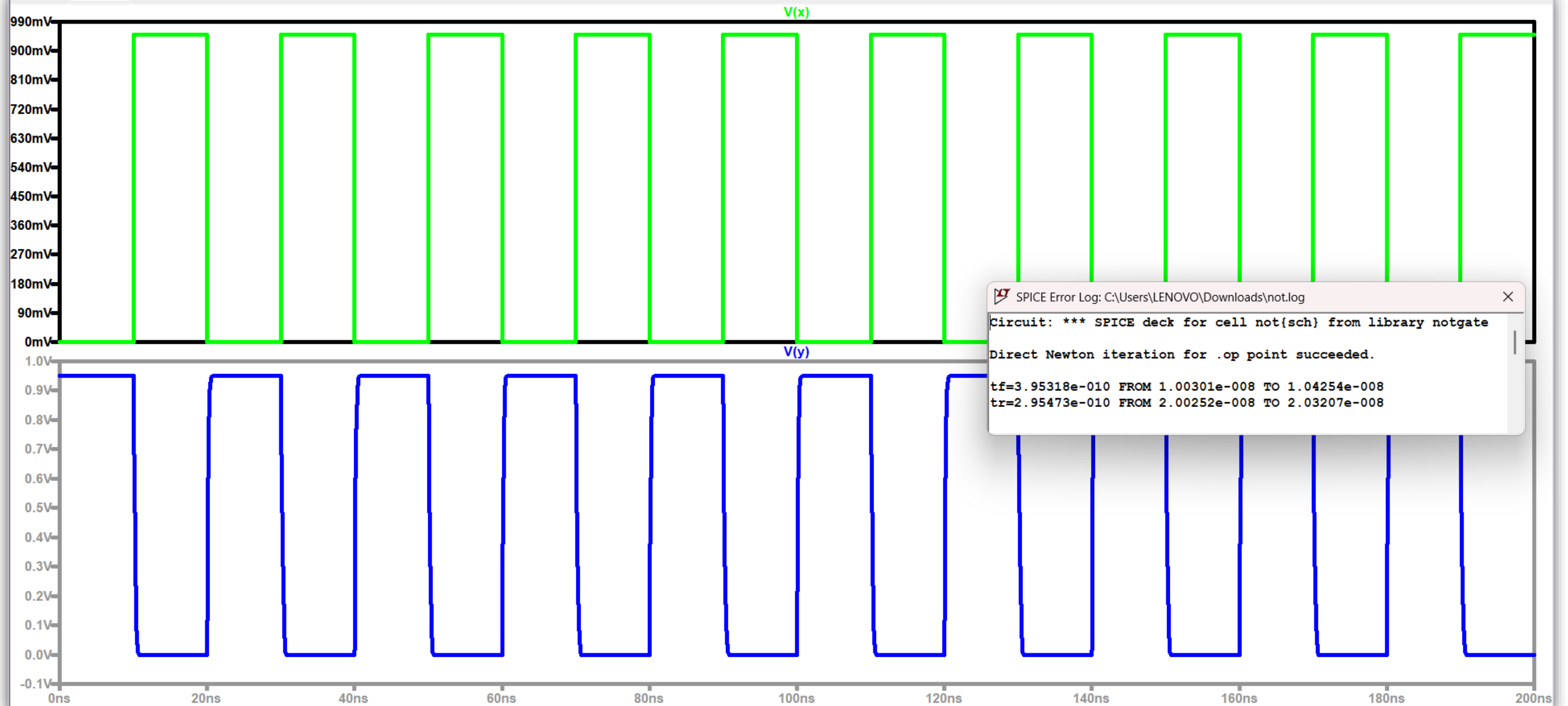
- 1- Initial tuning: Adjusted NOR gate and inverter sizes to equalize rise/fall times
- 2- Full circuit optimization: Applied sizing to complete schematic based on:
 - Minimizing D-to-Q delay
 - Achieving target power consumption
 - Meeting setup/hold time requirements

Inverter gate sizes

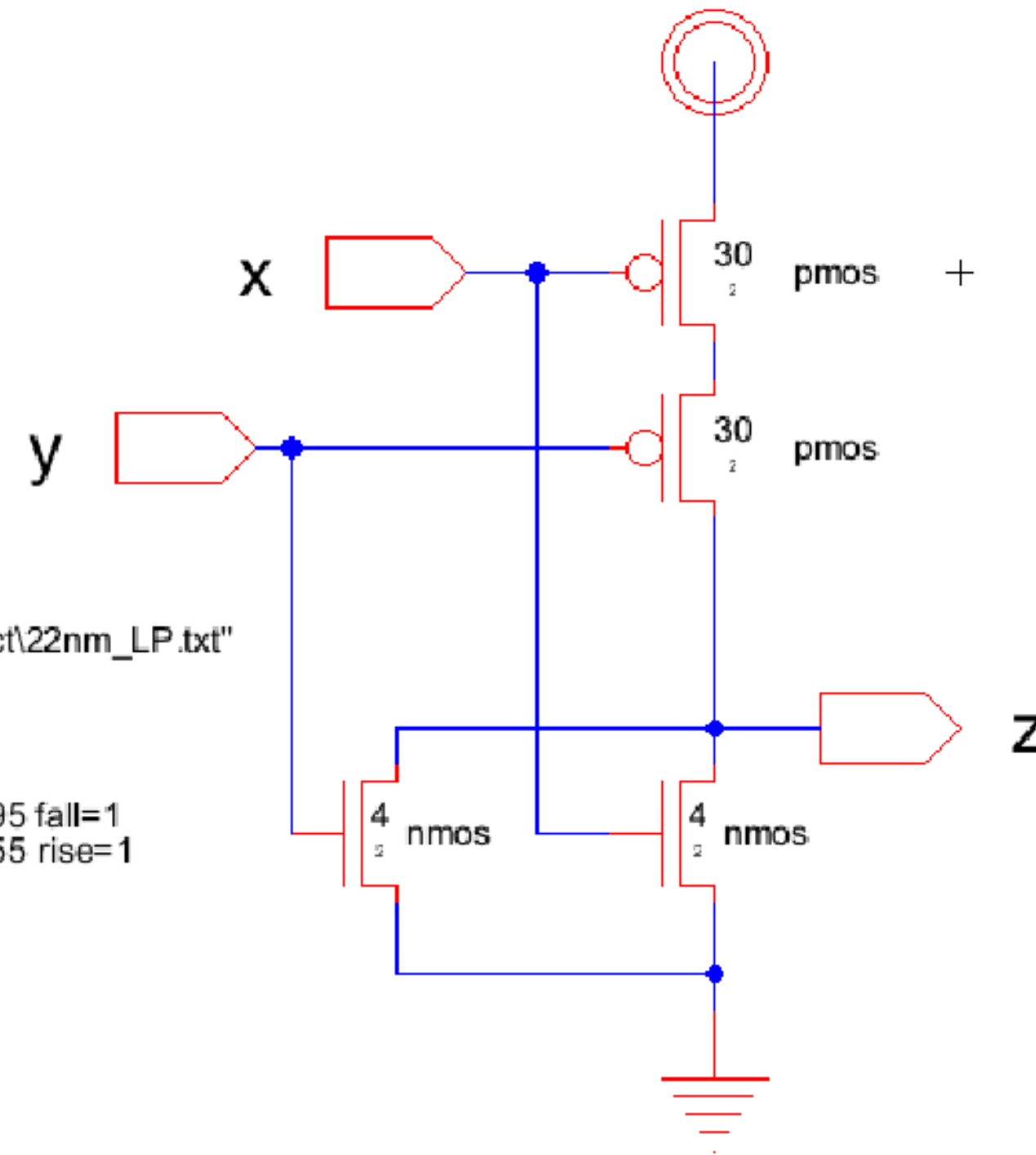
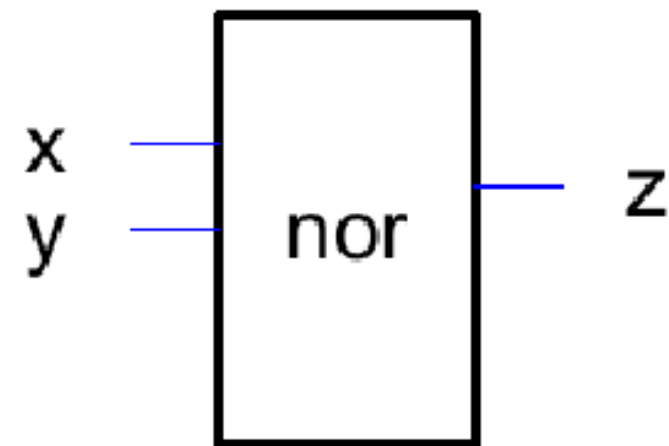
```
.include "C:\Users\LENOVO\Desktop\courses\fourth year\ic\project\22nm_LP.txt"
VDD VDD 0 0.95
VIN x 0 PULSE(0 0.95 10n 5p 5p 10n 20n)
CLOAD y 0 5f
.tran 1p 200n
.measure tran tf trig v(y) val=0.855 fall=1 targ v(y) val=0.095 fall=1
.measure tran tr trig v(y) val=0.095 rise=1 targ v(y) val=0.855 rise=1
.end
```



Inverter gate simulation

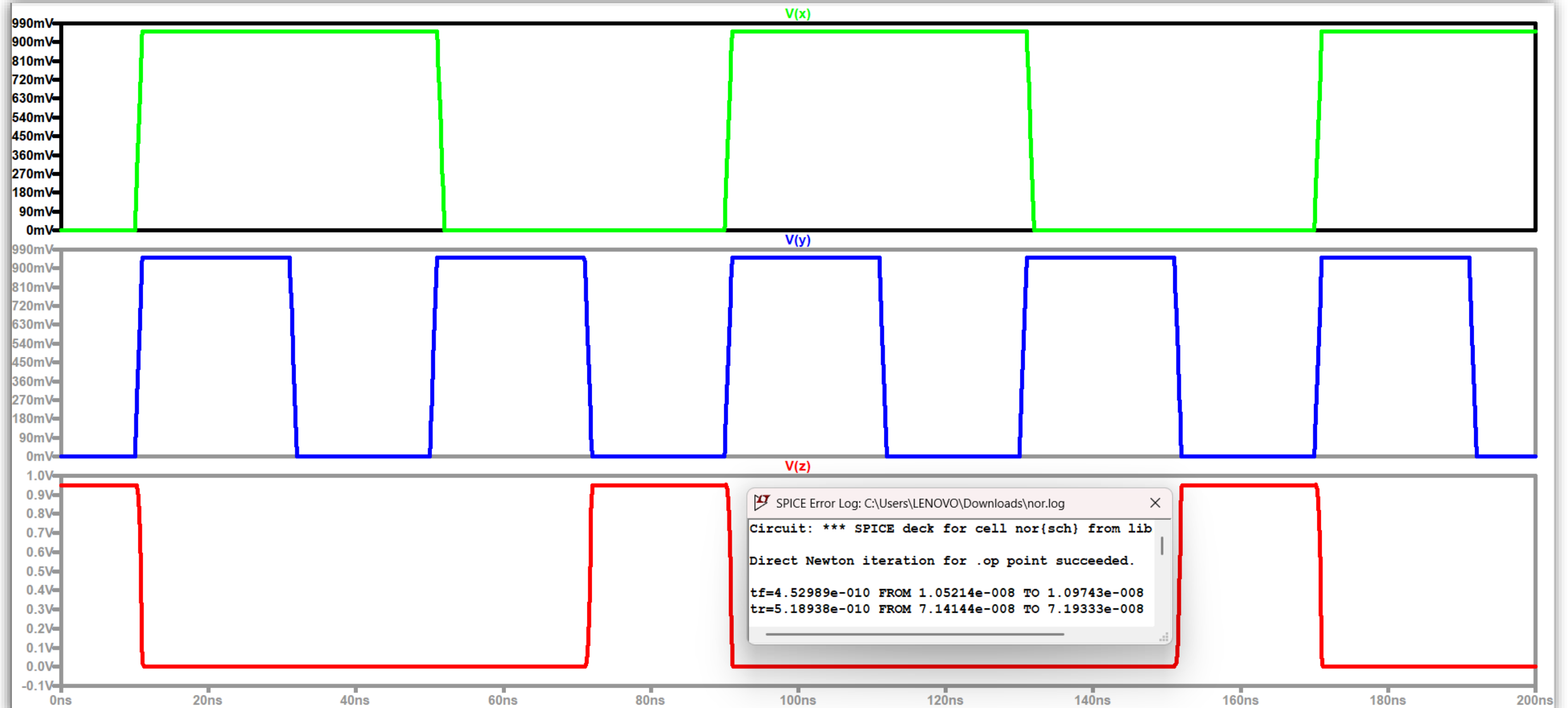


NOR gate sizes

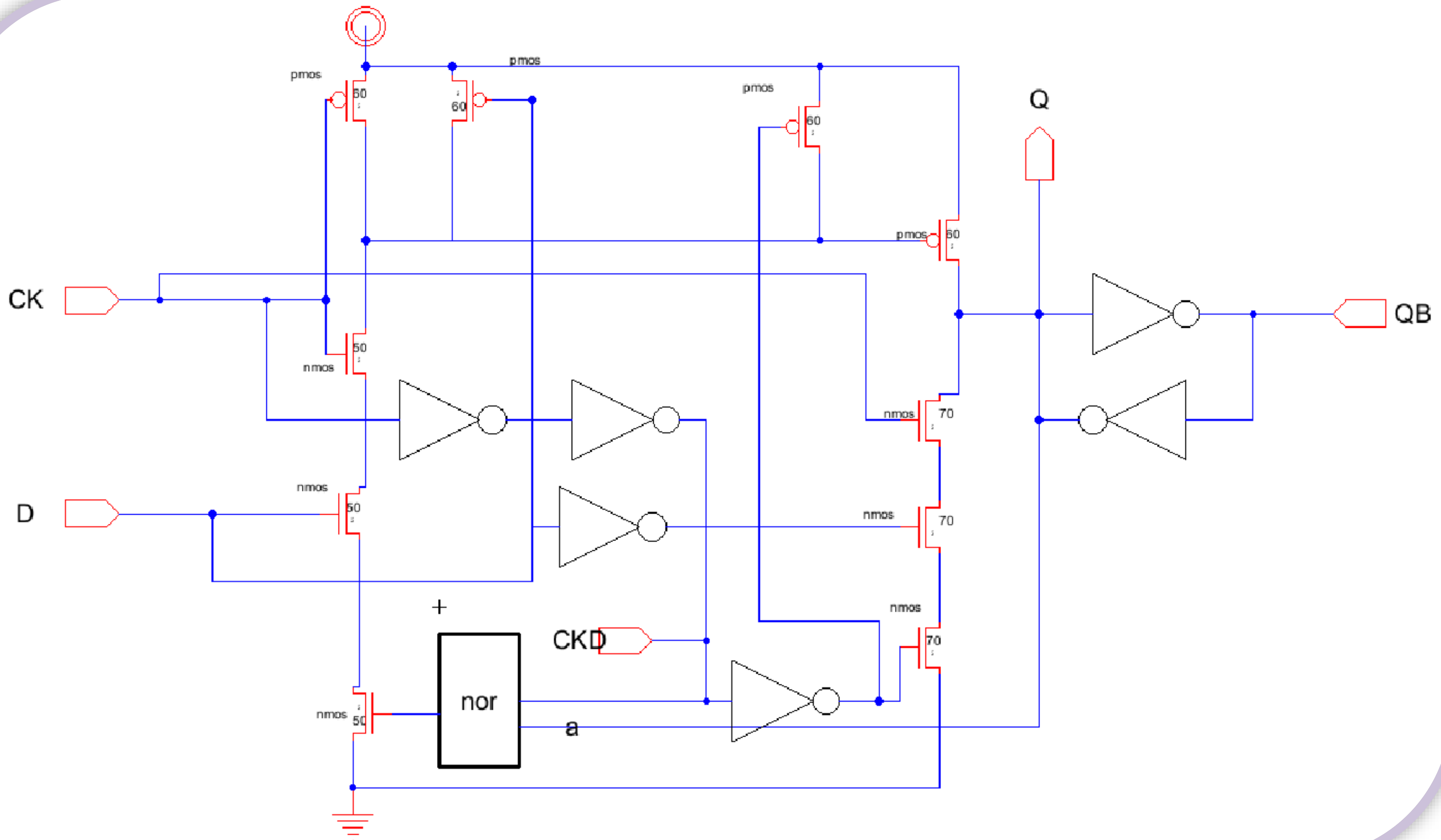


```
.include "C:\Users\LENOVO\Desktop\courses\fourth year\ic\project\22nm_LP.txt"
VDD VDD 0 0.95
vin1 x 0 pulse(0 0.95 10n 1n 1n 40n 80n)
vin2 y 0 pulse(0 0.95 10n 1n 1n 20n 40n)
load z 0 5f
.measure tran tf trig v(z) val=0.855 fall=1 targ v(z) val=0.095 fall=1
.measure tran tr trig v(z) val=0.095 rise=1 targ v(z) val=0.855 rise=1
.tran 10ps 200ns
```

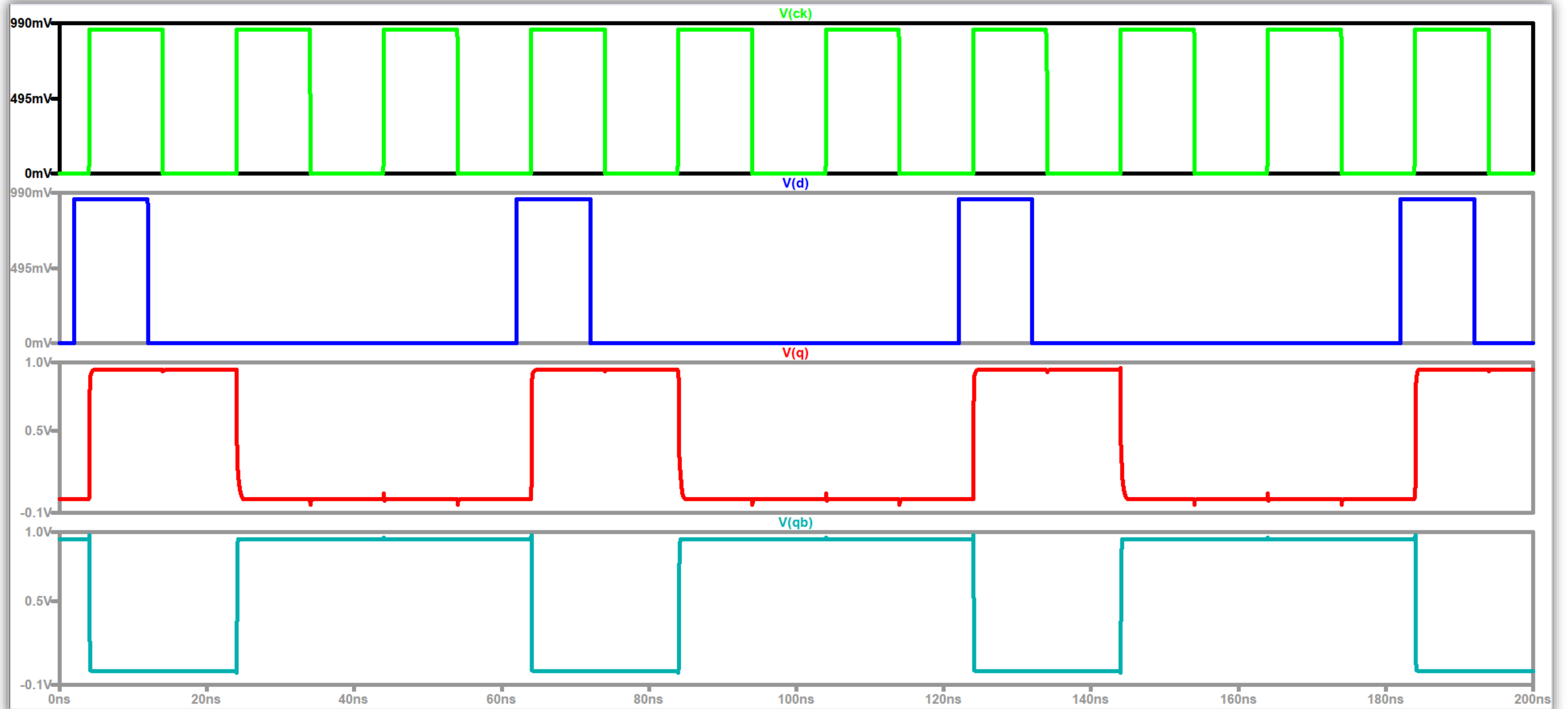
NOR gate simulation



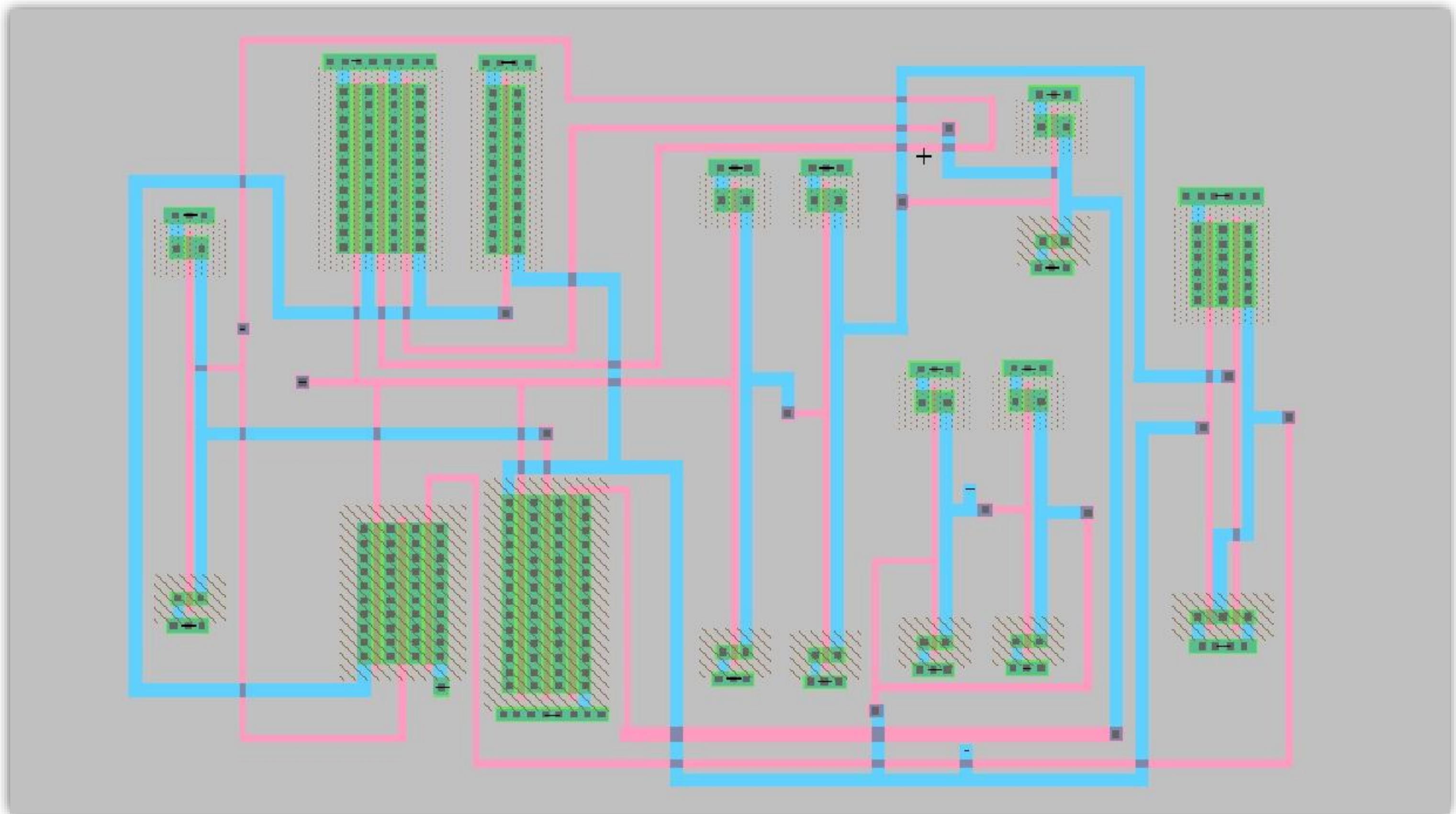
Single-Ended Schematic



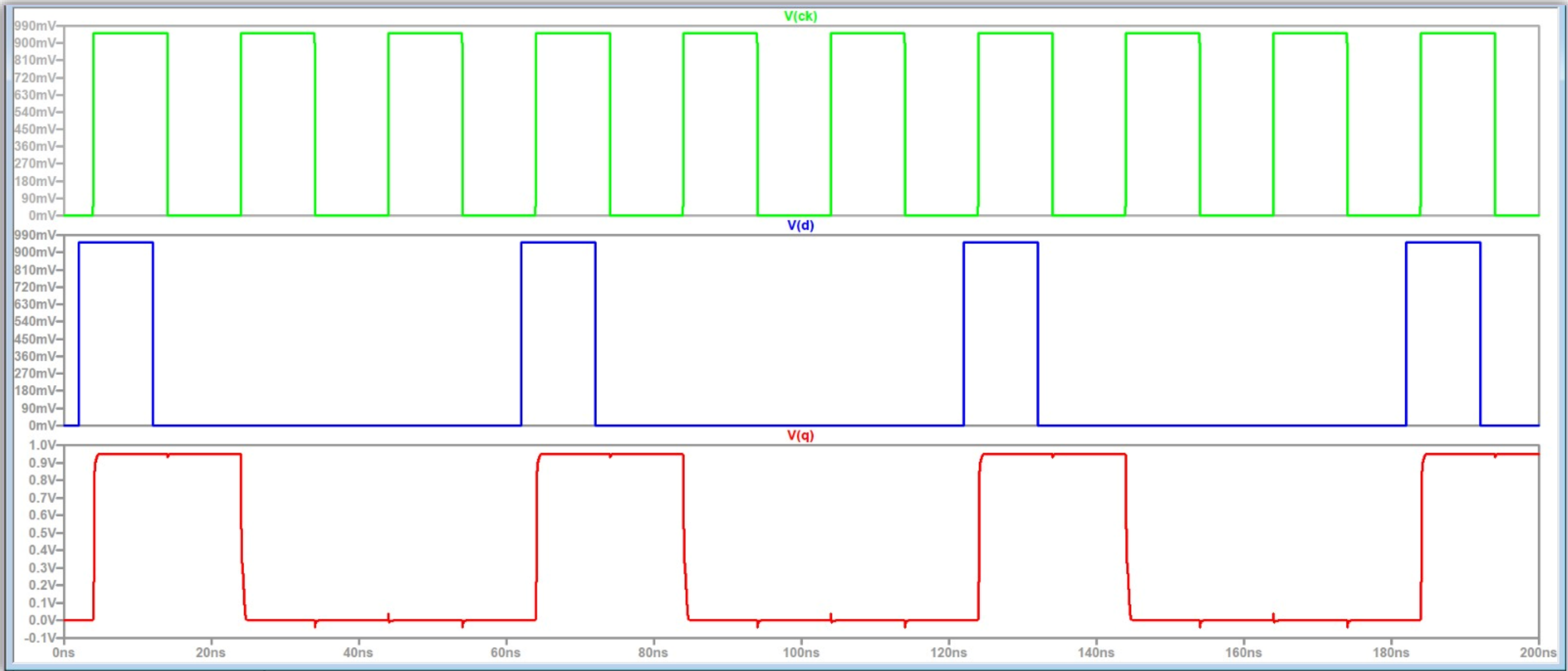
Simulation Results



Single-Ended Schematic



Simulation Results



COMPARISON BETWEEN PRE AND POST LAYOUT

Metric	Pre-Layout	Post-Layout
Clk-to-Q Delay	47.30 ps	52.03 ps
Total Power	1.96 μ W	2.33 μ W
Clock Power	0.154 μ W	0.166 μ W
Data Path Power	1.80 μ W	2.16 μ W
PDP	0.093 fJ	0.121 fJ
Setup Time	-200 ps	-180 ps
Hold Time	230 ps	260 ps

Implementation in 22nm CMOS

- ❑ **Layout Dimensions:** The core area of the Flip-Flop measures 518.5 lambda in length and 265 lambda in width.
- ❑ **Technology Scale:** With a scale factor of 8.0 nm per unit
- ❑ **Total Silicon Area:** The total estimated area is $8.8 \mu\text{m}^2$, demonstrating a highly compact implementation suitable for high-density VLSI systems.

Comparison Table

Design	Source	Technology	Supply (V)	Device Count	Clk-to-Q (ps)	Total Power (μW)	Clock Power (μW)	PDP (fJ)	Setup Time (ps)	Hold Time (ps)
Our Design		22nm	0.95	26	47.30	1.960	0.150	0.093	-200	230
CDFF (Proposed)	Paper 1	180nm	1.80	28	185.00	20.200	-	3.740	-	-
ep-DCO	Paper 1	180nm	1.80	26	184.00	24.400	-	4.490	-	-
CPFF	Paper 1	180nm	1.80	23	189.20	22.400	-	4.240	-	-
CCFF	Paper 1	180nm	1.80	26	206.00	22.600	-	4.660	-	-
DE-CPFF	Paper 1	180nm	1.80	33	226.00	21.600	-	4.880	-	-
imCCFF	Paper 1	180nm	1.80	32	233.00	27.000	-	6.290	-	-
SAFF_CP	Paper 1	180nm	1.80	24	545.00	21.800	-	11.880	-	-
CCFF (Single)	Paper 2	250nm	2.50	26	356.00	272.000	-	97.000	-10.00	230.00
SDFF	Paper 2	250nm	2.50	23	361.00	371.000	-	134.000	-	-
HLFF	Paper 2	250nm	2.50	20	365.00	351.000	-	128.000	-55.00	250.00
TGFF	Paper 2	250nm	2.50	18	433.00	283.000	-	123.000	-	-
SAFF (Diff)	Paper 2	250nm	2.50	28	332.00	467.000	-	155.000	-30.00	180.00
CCFF (Diff)	Paper 2	250nm	2.50	35	337.00	322.000	-	109.000	-50.00	200.00

PAPER 1 COMPARISON (180nm Technology)

Design	Their Power	Reduction %	Improvement Factor
CDFF (Proposed)	20.2 μ W	90.3%	10.3×
CPFF	22.4 μ W	91.3%	11.4×
CCFF	22.6 μ W	91.3%	11.5×
ep-DCO	24.4 μ W	92.0%	12.5×
DE-CPFF	21.6 μ W	90.9%	11.0×
SAFF_CP	21.8 μ W	91.0%	11.1×
imCCFF	27.0 μ W	92.7%	13.8×

PAPER 2 COMPARISON - Single-Ended (250nm Technology)

Design	Their Power	Reduction %	Improvement Factor
CCFF (Single)	272 μ W	99.3%	138.9×
SDFF	371 μ W	99.5%	189.5×
HLFF	351 μ W	99.4%	179.3×
TGFF	283 μ W	99.3%	144.6×

PAPER 2 COMPARISON - Differential (250nm Technology)

Design	Their Power	Reduction %	Improvement Factor
SAFF (Diff)	467 μ W	99.6%	238.5×
CCFF (Diff)	322 μ W	99.4%	164.5×